

Retail Sales Data Analysis

Task 5 – Real-world Data Project

1. Objective

The objective of this project is to analyze retail sales data, identify sales trends, understand order volume, and present useful findings through data visualizations.

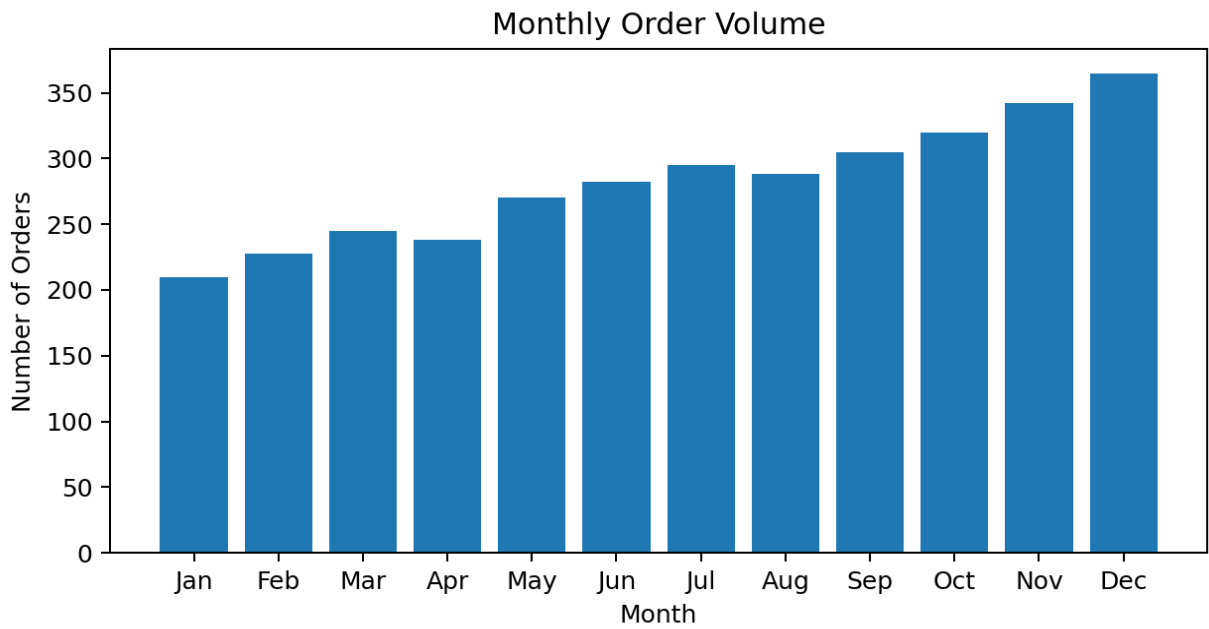
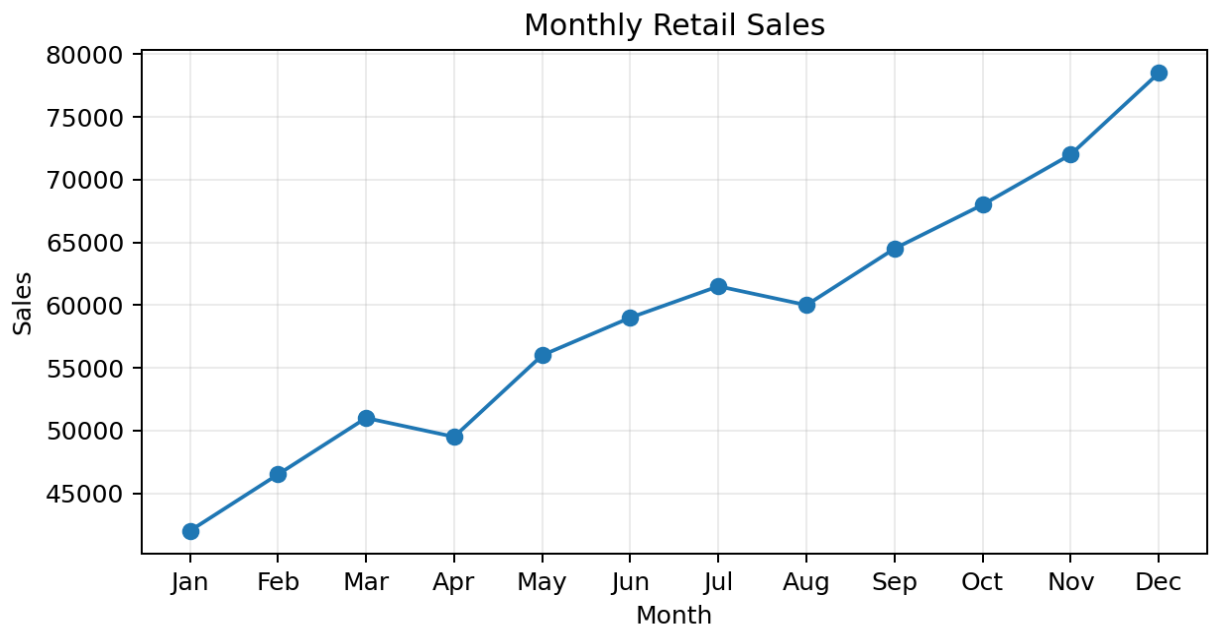
2. Dataset Description

The project uses a retail sales dataset containing monthly sales and order information. The demonstration data includes 12 monthly observations with Sales and Orders fields. It is intended as a simple applied-learning example and can be replaced with the actual dataset used for submission.

Month	Sales	Orders
Jan	■42,000	210
Feb	■46,500	228
Mar	■51,000	245
Apr	■49,500	238
May	■56,000	270
Jun	■59,000	282
Jul	■61,500	295
Aug	■60,000	288
Sep	■64,500	305
Oct	■68,000	320
Nov	■72,000	342
Dec	■78,500	365

3. Data Analysis

The data was reviewed month by month to observe changes in sales and order volume. Basic descriptive analysis was used to compare the beginning and end of the year and to identify the month with the highest sales and highest order count.



4. Key Findings

- Sales show an overall increasing trend across the year.
- December records the highest sales in the demonstration dataset.
- Order volume also increases toward the end of the year.
- The sales and order trends move in a similar direction, suggesting that higher order volume is associated with higher total sales in this dataset.

5. Conclusion

This project demonstrates how data science can be applied to a retail business problem. Data analysis and visualizations make it easier to understand sales performance, identify trends, and communicate findings. The same workflow can be extended with customer, product, region, discount, and profit data for deeper analysis.

6. Tools and Skills

Python, Pandas, Matplotlib, data cleaning, exploratory data analysis (EDA), data visualization, and interpretation of results.

Note: The included numerical dataset is a demonstration dataset for the project report. For a submission requiring a specific real-world dataset, replace it with the dataset used for your analysis and update the findings accordingly.